

**The University of Hong Kong**  
**Department of Geography**

**GEOG7305 Programming for Geospatial Data (6 credits)**

**Instructors**

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Classroom: LE1

Office hours:

**Objectives**

To equip students with the skills necessary to design and implement a GIS system that is capable of addressing a specific set of application issues.

**Course synopsis**

This course is a hands-on course that provides students with practical experience in designing and implementing GIS projects. Students will learn the basic principles of GIS project design and work independently on a project that is relevant to their field of interest. Topics covered in the course include GIS system design, database design, Python programming, and workflow diagram design. Students will gain extensive experience in using and customizing GIS software and learn how to design and implement a GIS system that meets the specific needs of their project through Python programming. Additionally, students will also learn how to design workflow diagrams, which are an essential tool for organizing and visualizing complex GIS projects. Throughout the course, students will apply the GIS techniques they have learned in a substantive and meaningful way by working on a self-designed project. Students will be expected to demonstrate competence in using and developing GIS techniques, and to produce a final project that showcases their skills and knowledge.

**Lecture topics**

- Overview of GIS and GIS application system design
- Database design
- Python basics
- Python code editing and debugging
- Constructing workflows using ModelBuilder
- Working with layers, tables, geometries and rasters
- Manipulating spatial data with Python code

**Course Assessment**

60% Coursework + 40% Examination

*Details of coursework components*

- *Lab reports (20%): Lab reports will be submitted periodically throughout the semester.*
- *Group project (40%): students will be divided into groups, with each group developing a comprehensive case study on a chosen topic that related to any implementation and customization of GIS. Students are required to present their project and write up a project report. **The submission deadline for the project report is May 8, 2026.***

***The final examination for this course is scheduled to take place on May 15, 2026.***

### **Lecture overview**

| <b>Week</b>                                       | <b>Schedule</b>                                                                         | <b>Date</b>                       |
|---------------------------------------------------|-----------------------------------------------------------------------------------------|-----------------------------------|
| 1                                                 | Lecture: GIS system design and analysis                                                 | 23 January                        |
| 2                                                 | Lecture: Database design                                                                | 30 January                        |
| 3                                                 | Lecture: Basics of Python<br>Lab: DFD Diagram of Smart City/Environment                 | 6 February                        |
| 4                                                 | Lecture: GIS customization<br>Lab: ER Diagram                                           | 13 February<br>(Lab 1 report due) |
| Class Suspension for Chinese New Year Feb 17 - 23 |                                                                                         |                                   |
| 5                                                 | Lecture: GIS customization: dataset and layer management<br>Lab: Introduction to python | 27 February<br>(Lab 2 report due) |
| 6                                                 | Lecture: GIS customization: attribute data management<br>Lab: Modelbuilder              | 6 March<br>(Lab 3 report due)     |
| 7                                                 | (Reading Week)                                                                          | 13 March                          |
| 8                                                 | Mid-term Project Presentation                                                           | 20 March<br>(Lab 4 report due)    |
| 9                                                 | Lecture: GIS customization: raster data operations<br>Lab: Python for geoprocessing     | 27 March                          |
| 10                                                | Public Holiday                                                                          | 3 April                           |
| 11                                                | Lecture: GIS customization: spatial analysis<br>Lab: Selection query                    | 10 April<br>(Lab 5 report due)    |
| 12                                                | Lecture: GIS customization: suitability analysis                                        | 17 April<br>(Lab 6 report due)    |
| 13                                                | Final Project Presentation                                                              | 24 April                          |
| 14                                                | Public Holiday                                                                          | 1 May                             |

### **Course Learning Outcomes (CLOs)**

At the end of the course, students will achieve the following learning outcomes:

- Demonstrate a comprehensive understanding of the principles of GIS system design and implementation.
- Develop proficiency in using Python programming language to write scripts and automate GIS tasks.
- Design and implement workflows using a range of GIS tools and techniques.

- Customize and extend ArcGIS tools and workflows using Python to meet specific project requirements.
- Demonstrate competence in using and developing GIS techniques, and be able to apply them in a substantive and meaningful way to solve real-world problems.

### **Required readings**

Paul A. Zandbergen, 2020. *Python Scripting for ArcGIS Pro*, ESRI Press.

*Online materials:*

Python in ArcGIS Pro: <https://pro.arcgis.com/en/pro-app/latest/arcpy/get-started/installing-python-for-arcgis-pro.htm>